PROJECT PRESENTATION

Medicine Information AI Assistant

A single page web app that turns a medicine name into clear information on uses, doses, side effects, warnings and alternatives, with a built-in chatbot.

HTML | CSS | JavaScript | No installation

Walkthrough of the working website

THE IDEA

Medicine facts are scattered and hard to read

The problem

- Leaflets and websites bury the key facts in dense text.
- People want a quick answer: what is it for, how much, what to avoid.
- Warnings about pregnancy, alcohol or other medicines are easy to miss.

What this project does about it

- One search box leads to a structured card for each medicine.
- Information is split into tabs, so each question has one place to look.
- Scores, charts and a comparison tool make the data easier to read at a glance.
- A chatbot answers plain-language questions from the same data.

OVERVIEW

What the website can do

FIND

Smart search

Live autocomplete by name, generic name or category, plus voice input and popular medicine shortcuts.

LEARN

Seven-tab medicine card

Uses, side effects, dosage, warnings, safety tips, alternatives and composition for every medicine.

JUDGE

AI insights

Risk, safety, popularity and recommendation scores shown as gauges beside each medicine.

OVERVIEW

Analytics dashboard

Pie charts for categories and prescription status, and a table of every medicine in the database.

DECIDE

Compare two medicines

Side by side view of price, prescription status, dose limits and scores.

ASK

MediBot chatbot

Answers plain questions, remembers the last medicine, searches by symptom and flags emergencies.

UNDER THE HOOD

One data source feeds every screen

medicinesDatabase
A JavaScript array holding uses, doses, warnings and scores

→

Search
Matches name, generic name and category

→

Detail view
Medicine card and tabs generated from templates

→

Insights
Gauges, dashboard and comparison

MediBot reads the same array

It matches the question to a topic (dosage, side effects, alcohol and so on) and builds the answer from the stored fields, so chat and screens never disagree.

Saved in the browser

Recent searches, favorites and the light or dark theme are kept in the browser's local storage. No account or server is needed.

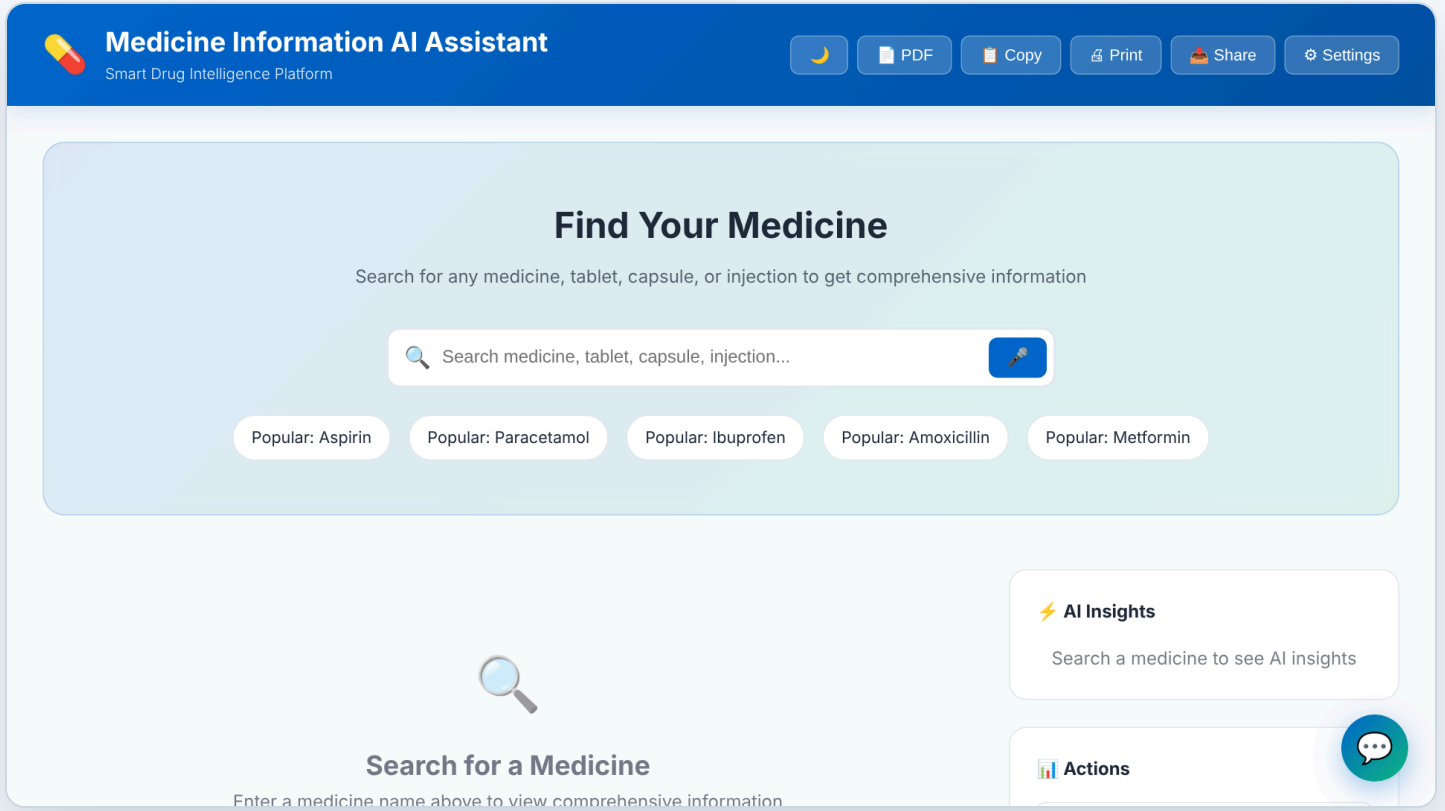
HTML | CSS variables and dark mode | Vanilla JavaScript | Canvas charts

SCREEN 1

The home screen

Everything starts from one search box, with tools one click away.

- **Header buttons:** dark mode, PDF, copy, print, share and settings.
- **Search area:** type or speak a medicine name.
- **Popular shortcuts:** one click opens Aspirin, Metformin and others.
- **Sidebar:** AI insights, actions, recent searches and favorites.



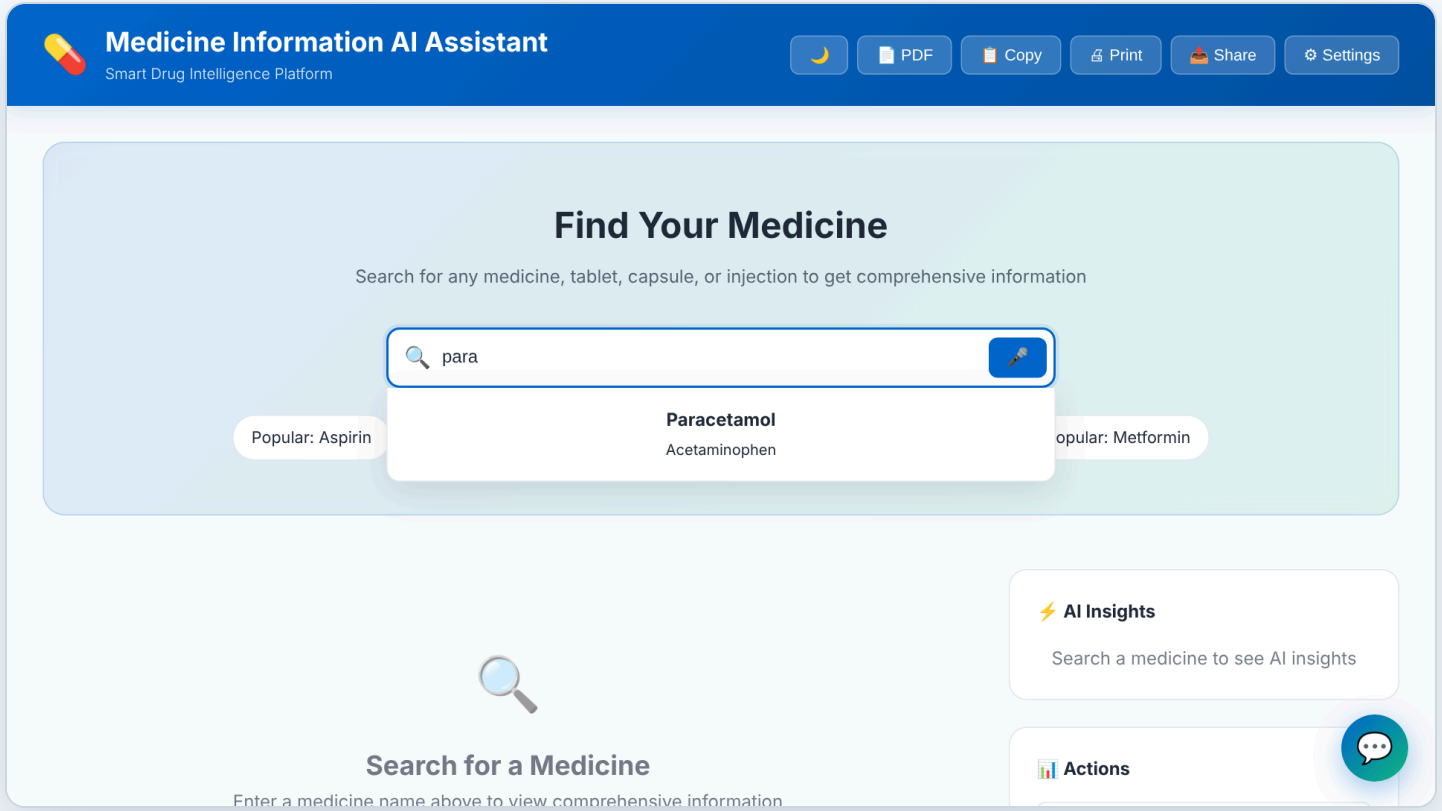
The empty state invites a search; insights fill in once a medicine is chosen.

SCREEN 2

Search with live autocomplete

Suggestions appear as you type, so a partial name is enough.

- Typing **"para"** suggests Paracetamol with its generic name, Acetaminophen.
- Matches brand, generic name and category.
- The microphone button lets you speak the name instead.



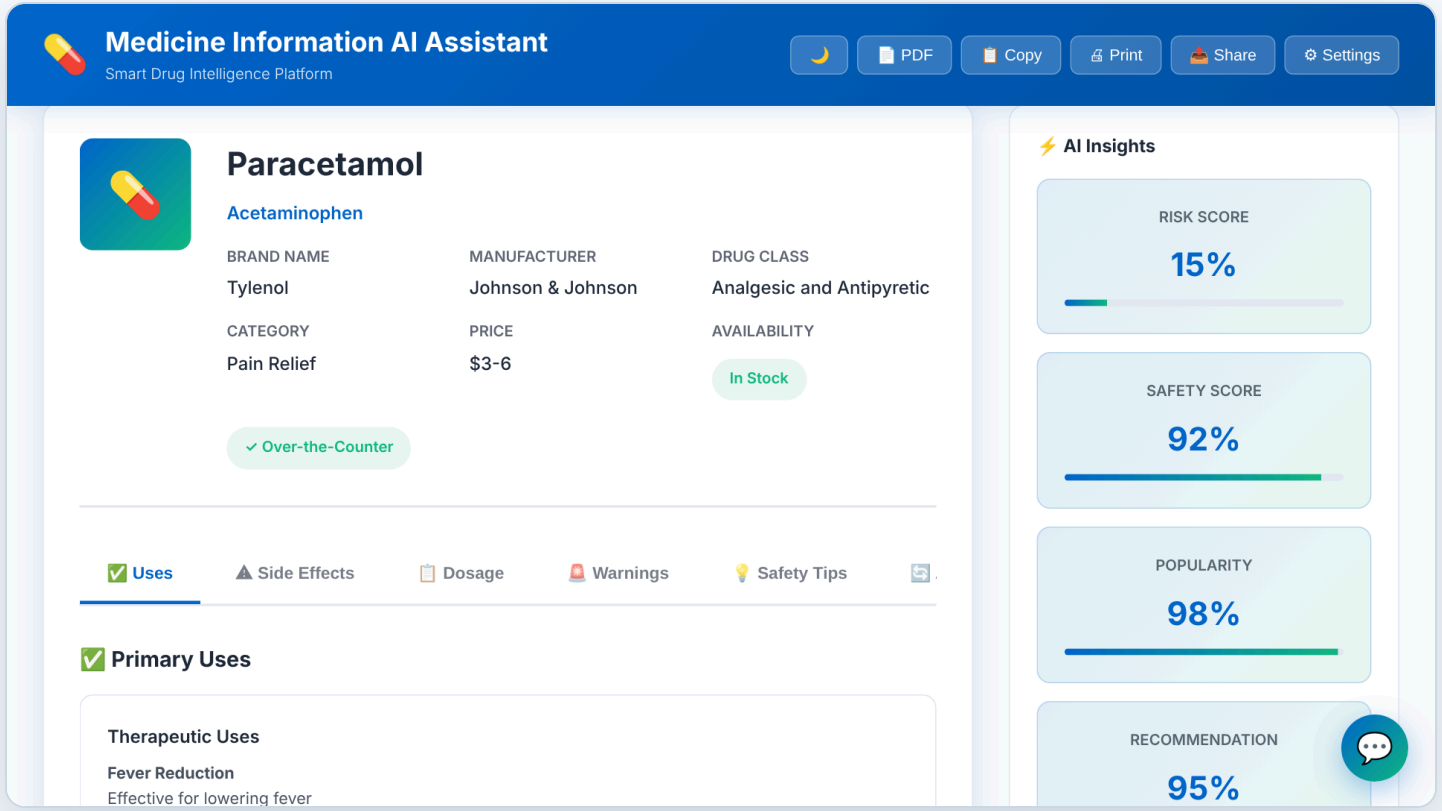
Autocomplete dropdown under the search bar.

SCREEN 3

The medicine card and its uses

The top of the card answers the basics before you read anything else.

- Generic and brand name, manufacturer, drug class and category.
- Price range and stock status.
- A clear badge: **Over-the-Counter** or **Prescription Required**.
- Seven tabs below organise the details, starting with **Uses**.



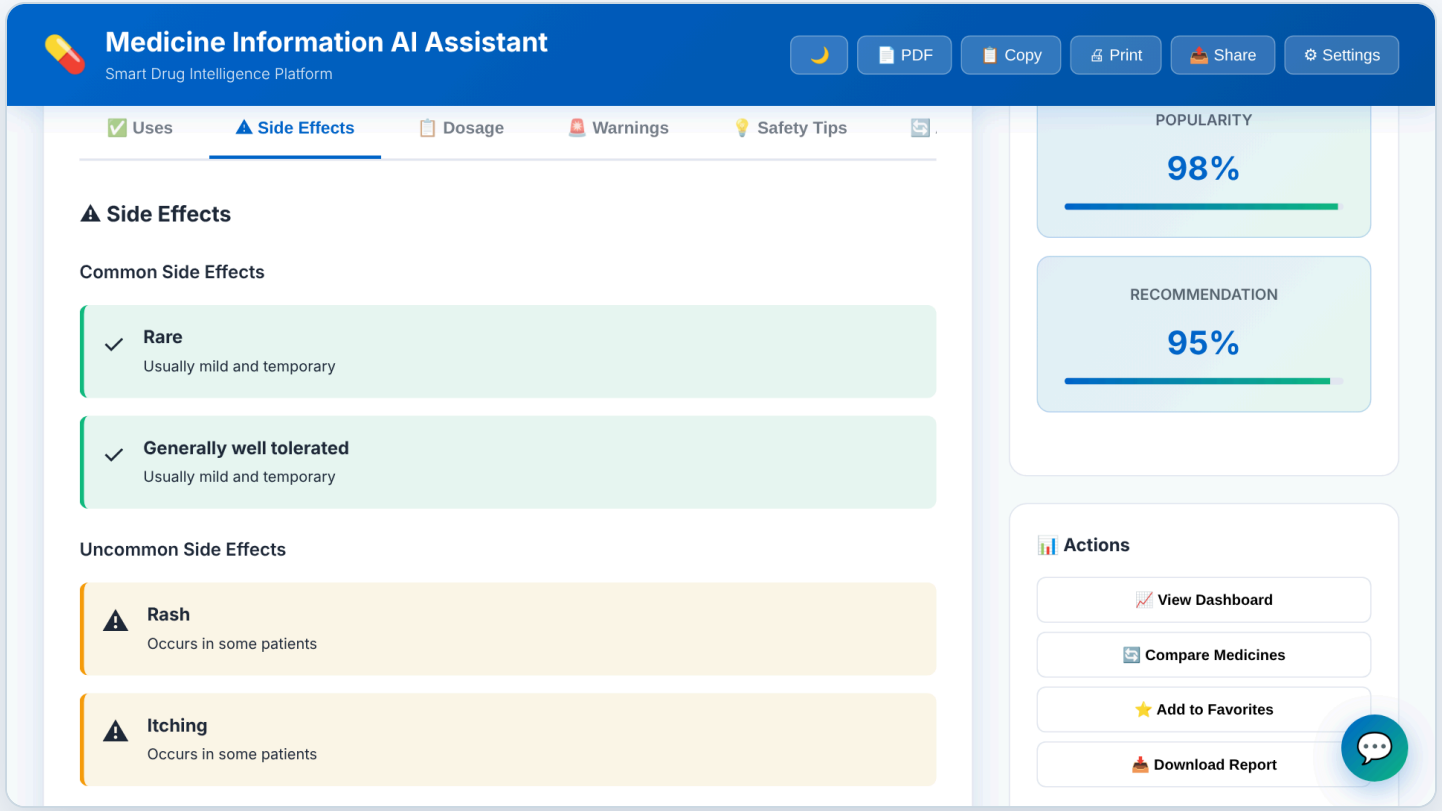
Paracetamol card with the Uses tab open.

TAB: SIDE EFFECTS

Side effects grouped by severity

Colour tells you how seriously to take each effect.

- **Green:** common and usually mild.
- **Amber:** uncommon, occurs in some patients.
- **Red:** serious, needs urgent medical attention.



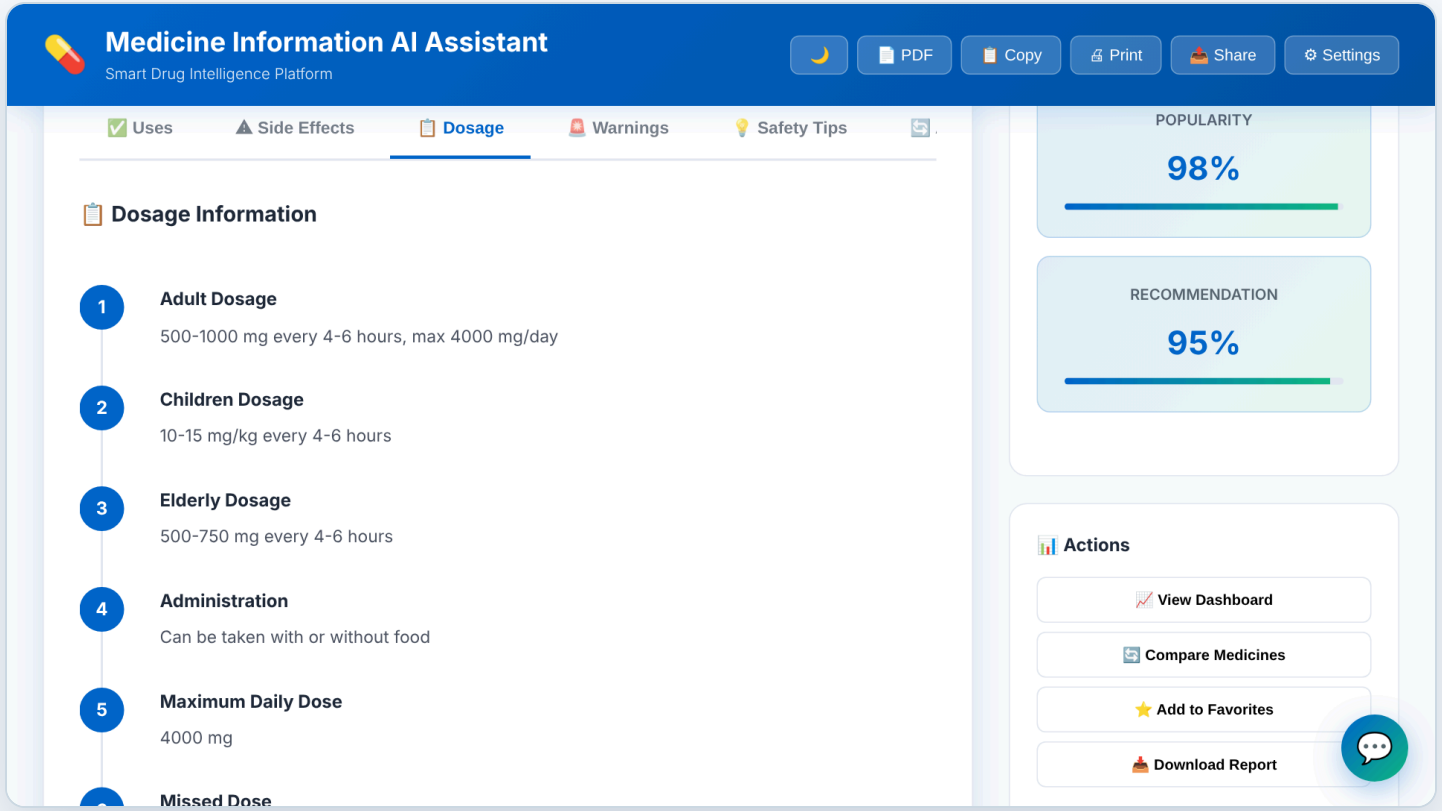
Common and uncommon side effects as separate colour-coded cards.

TAB: DOSAGE

A dosage guide for every age group

Everything about taking the medicine correctly, in one tab.

- Adult, child and elderly doses.
- How to take it and the maximum per day.
- What to do after a missed dose.
- What to do in case of overdose.



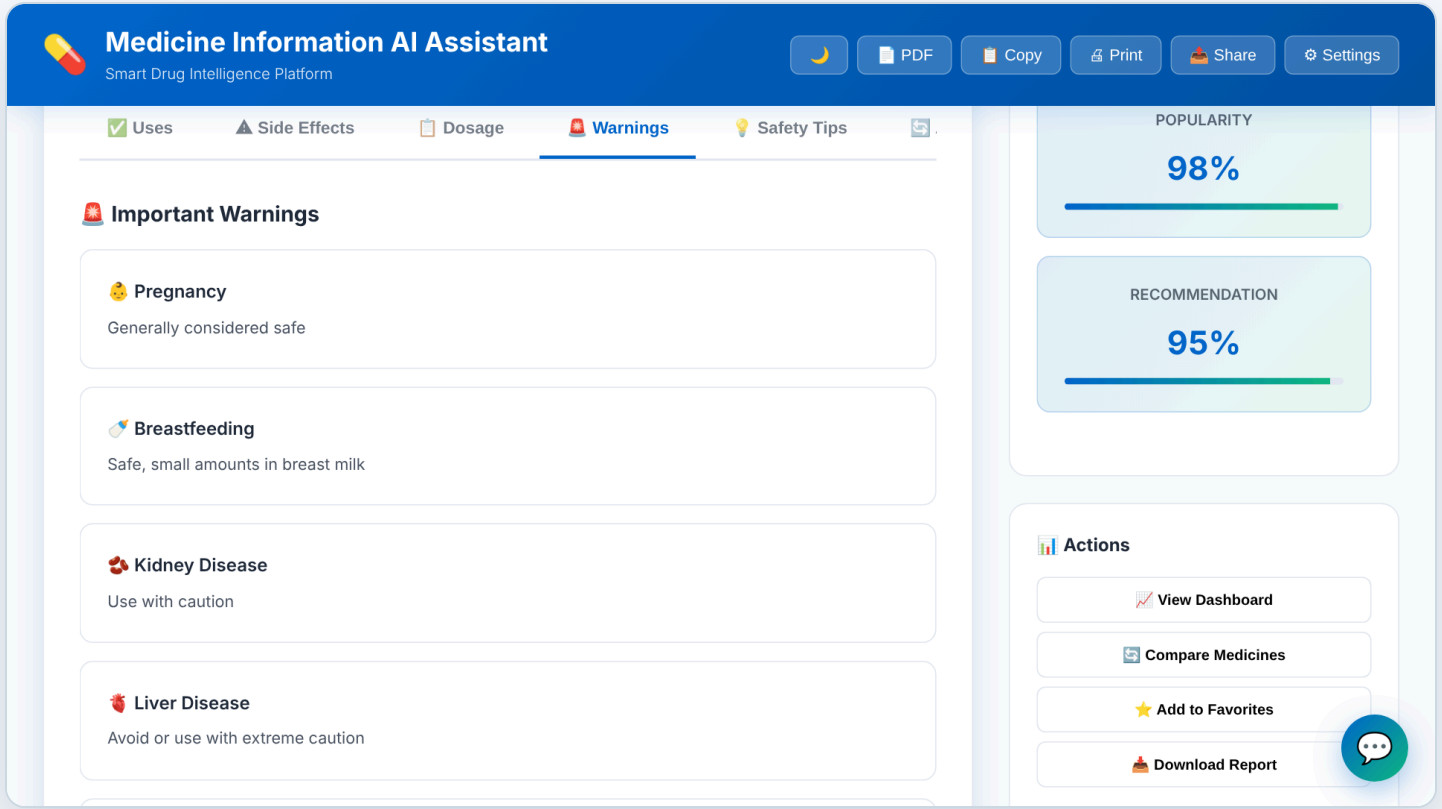
Doses are typical values from the database. A doctor's or label's instructions always come first.

TABS: WARNINGS AND SAFETY TIPS

Warnings and safety advice

The situations where a medicine needs extra care.

- Pregnancy and breastfeeding.
- Kidney and liver disease.
- Alcohol, driving and allergies.
- Interactions with other medicines.
- The Safety Tips tab adds storage and usage advice.



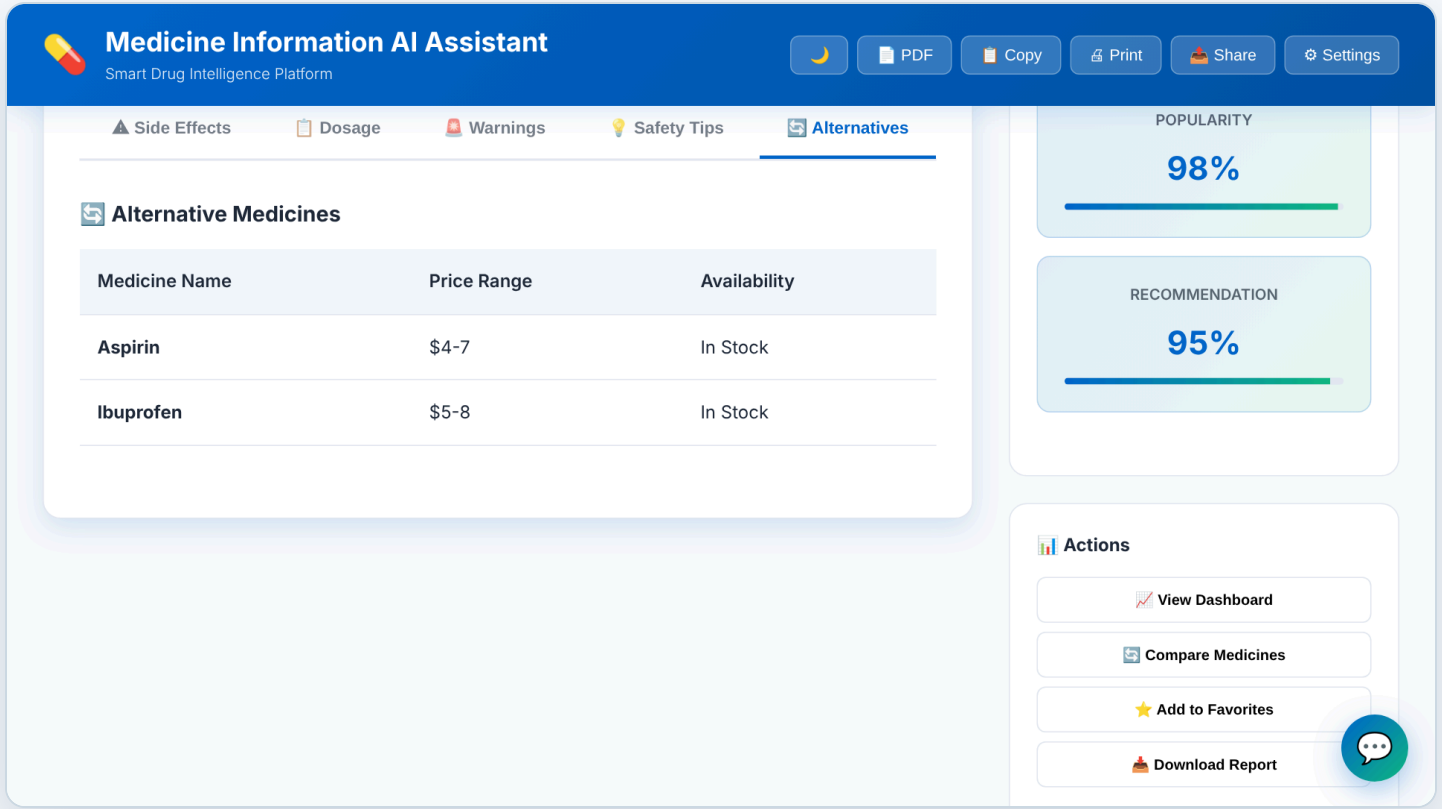
The Warnings tab.

TABS: ALTERNATIVES AND COMPOSITION

Alternatives and ingredients

What else could be used, and what is actually inside.

- Alternative medicines with price range and availability.
- Composition tab: active ingredient, strengths and forms.
- Also explains how the medicine works in the body.



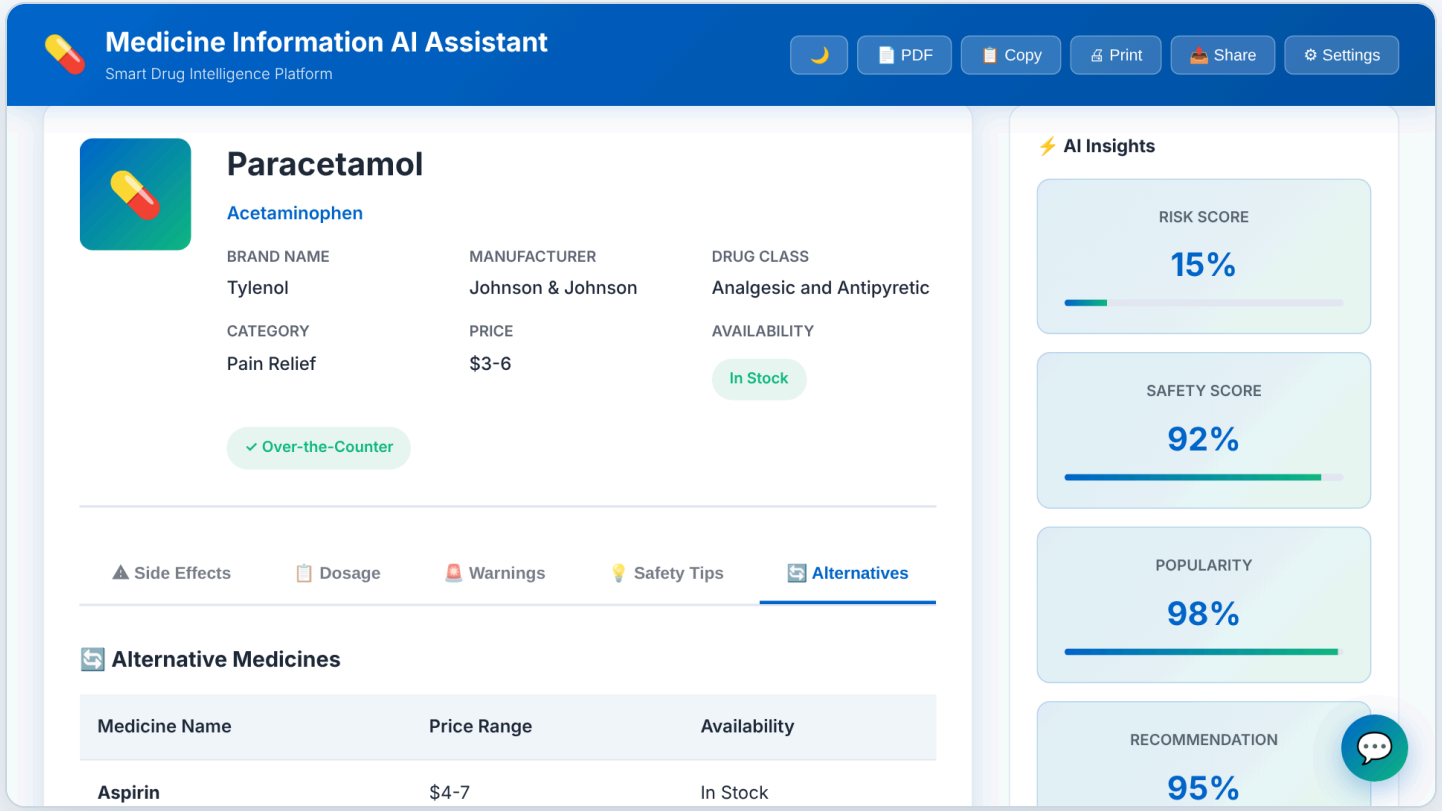
Alternatives table for Paracetamol.

SIDEBAR

AI insights at a glance

Four scores summarise each medicine as gauges.

- **Risk score:** lower is better.
- **Safety score:** overall safety rating.
- **Popularity:** how widely it is used.
- **Recommendation:** combined suggestion score.



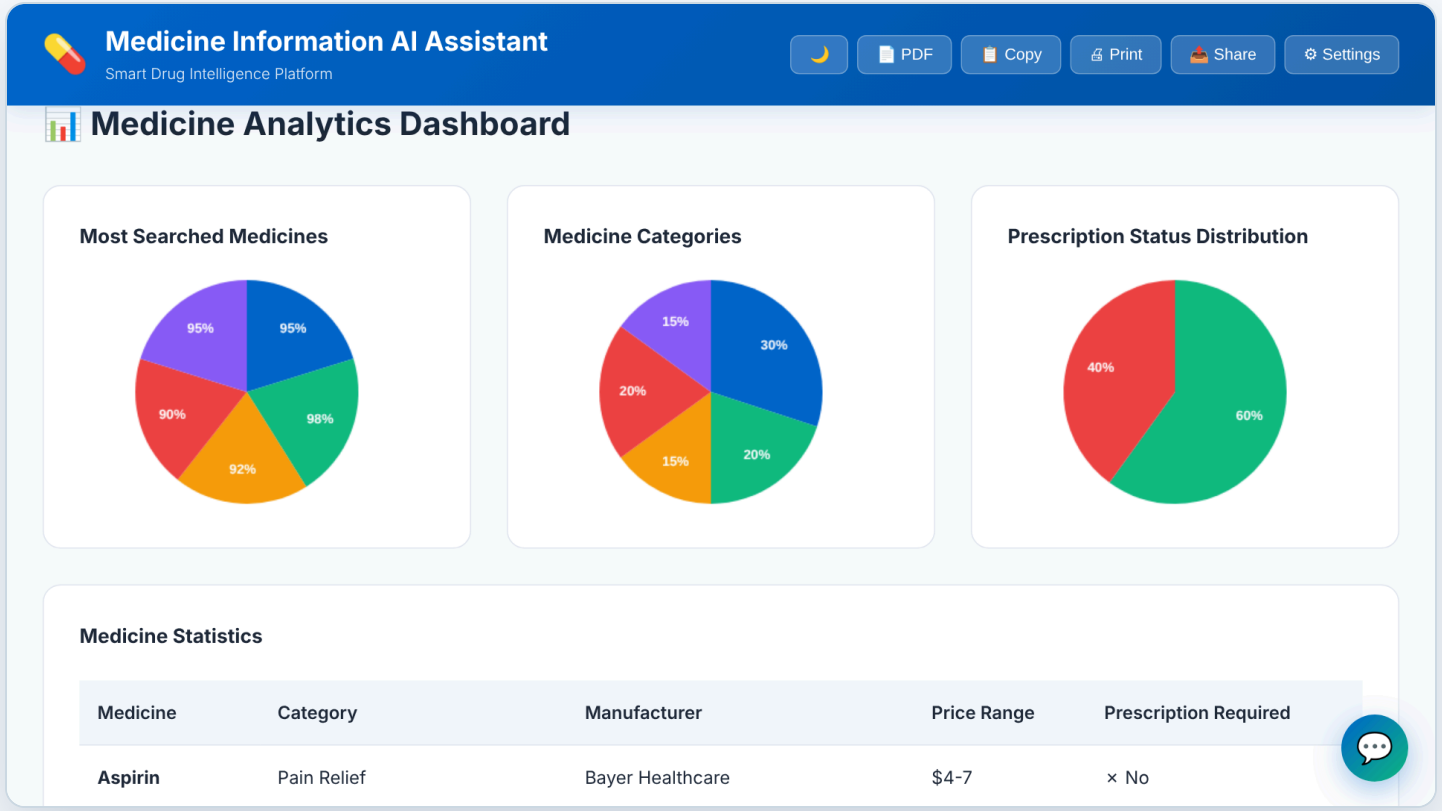
Paracetamol: 15% risk, 92% safety, 98% popularity. Scores come from the database, not from a clinical model.

DASHBOARD

Analytics dashboard

An overview of the whole database instead of one medicine.

- Pie charts: most searched, categories, prescription status.
- Table of every medicine with category, manufacturer, price and prescription need.
- Charts are drawn on canvas, with no chart library.



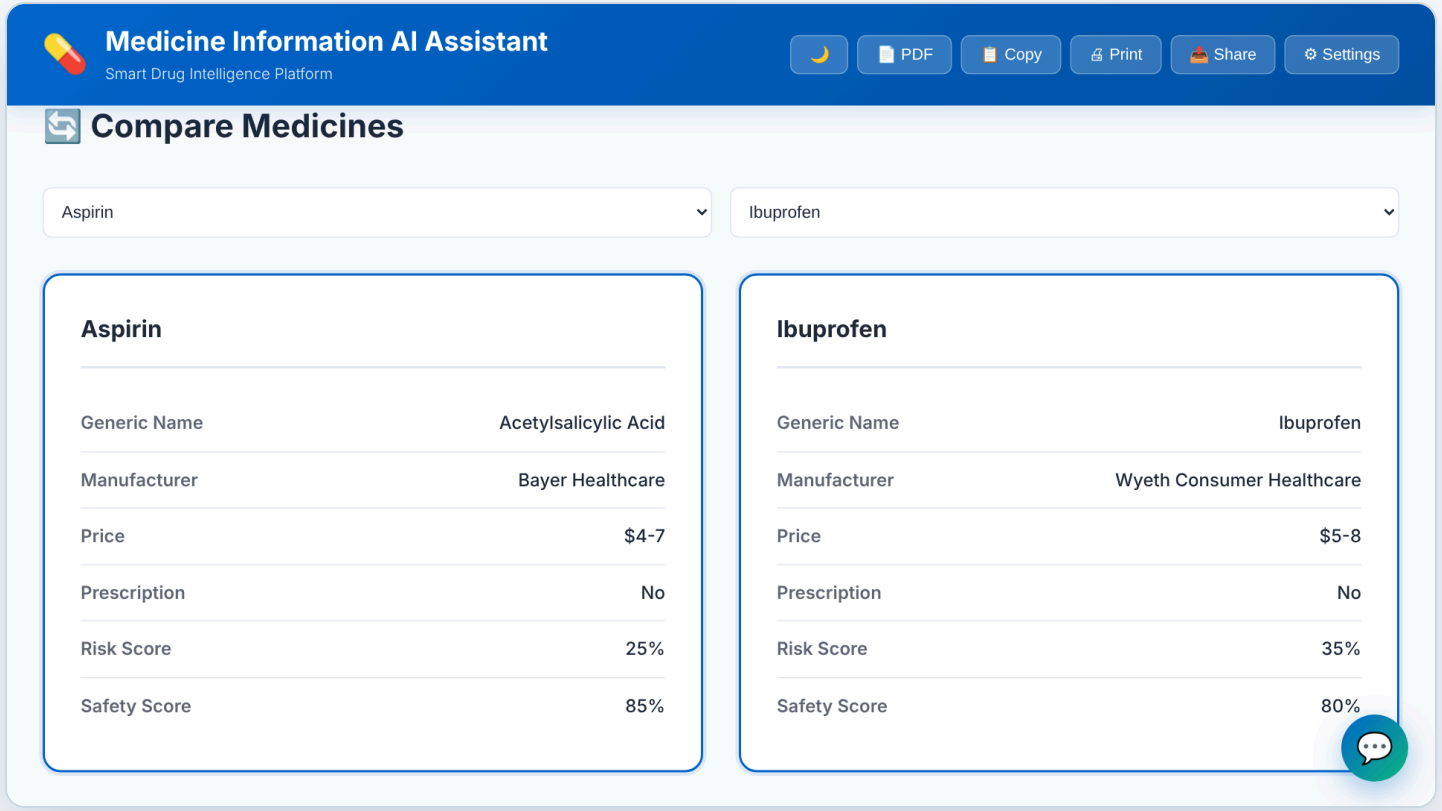
The chart figures are sample data for demonstration.

COMPARE

Compare two medicines side by side

Pick any two from the drop-downs to see the differences.

- Price and prescription status.
- Dose limits and drug class.
- Risk and safety scores.



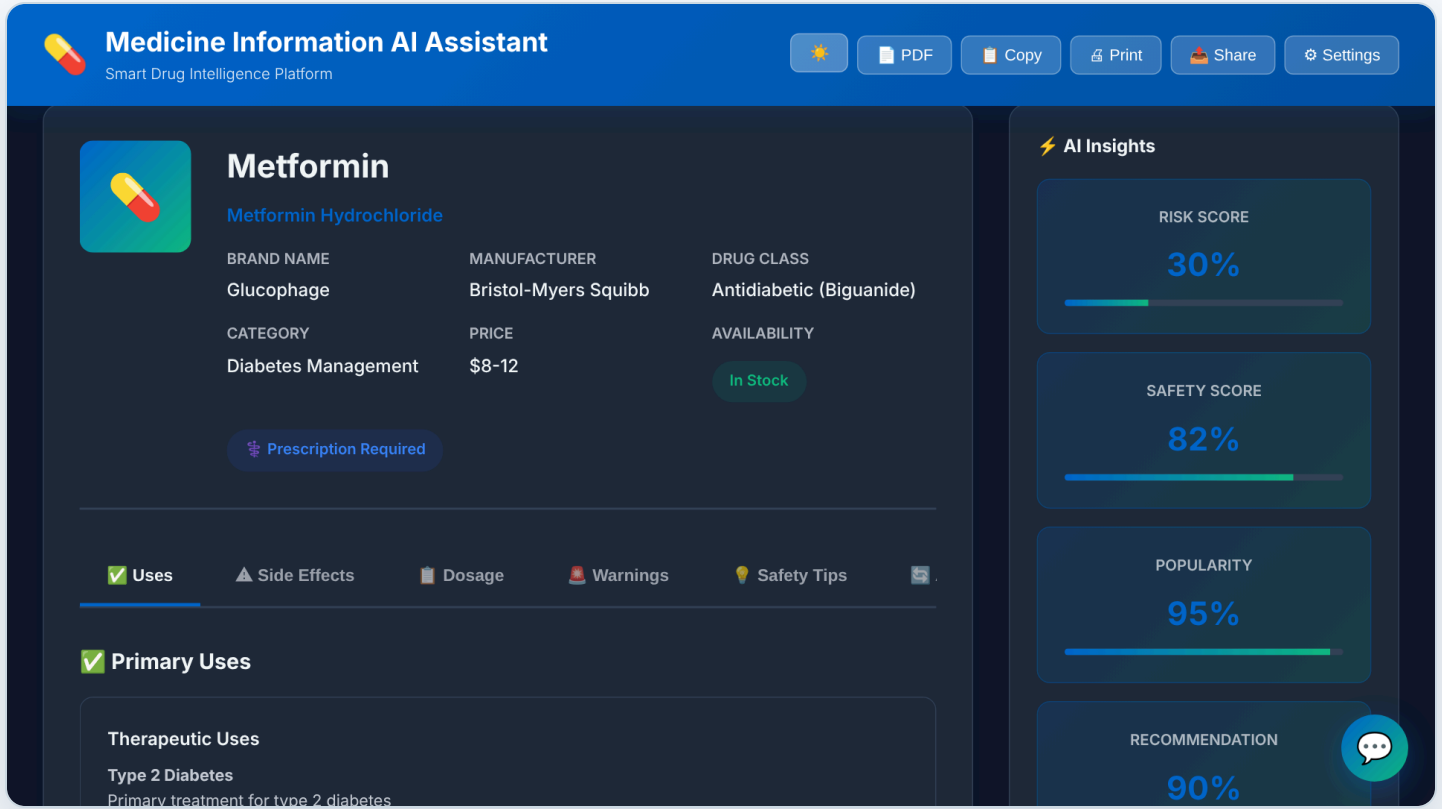
Example: Aspirin compared with Ibuprofen.

COMFORT AND CONVENIENCE

Dark mode and saved preferences

The page adapts to the user and remembers them.

- One click switches between light and dark themes.
- Recent searches, favorites and the chosen theme are saved in the browser.
- Everything is stored on the user's own device.



Metformin card in dark mode.

NEW FEATURE

MediBot: ask in plain language

A floating chat window answers from the same database.

- Understands topics such as dosage, side effects, alcohol, pregnancy and alternatives.
- Remembers the last medicine, so "what about side effects?" works.
- Compares two medicines and suggests medicines by symptom.
- Quick-reply buttons open the full details card.

How it works: a keyword and pattern matcher picks the topic and the medicine, then fills an answer template from the database. It is a rule-based assistant, so unusual wording may not be understood.



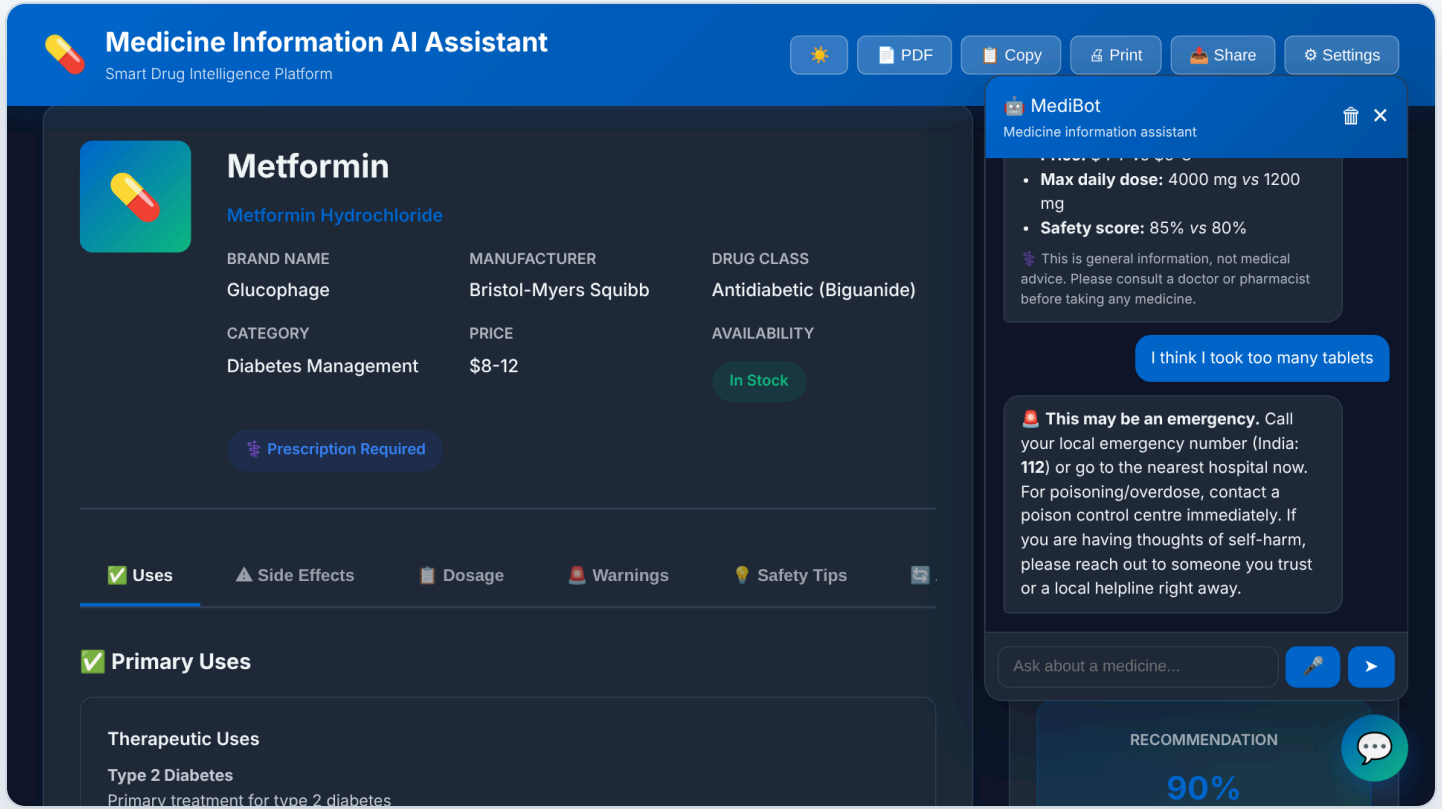
Left: a dosage question, then a follow-up. Right: comparing Aspirin and Ibuprofen.

SAFETY FIRST

Emergency detection

Some messages should never get a normal answer.

- Words such as overdose, chest pain, trouble breathing or self-harm trigger an urgent message.
- It points to the local emergency number (112 in India) and nearby hospital care.
- Every answer carries a reminder that this is general information, not medical advice.



Response to "I think I took too many tablets".

HONEST LIMITS AND NEXT STEPS

What to know, and where it could go

Current limits

- Data comes from a small built-in database and must be checked against official sources before real use.
- Dashboard figures are sample numbers.
- The chatbot is rule-based: it matches keywords and does not truly understand language.
- It gives general information and cannot replace a doctor or pharmacist.

Possible next steps

- Connect a verified drug database or API for more medicines.
- Upgrade the chatbot to a language model through a small backend, using the database as its source.
- Add more languages and a mobile-friendly layout polish.
- Add optional user accounts so favorites follow the user across devices.

SUMMARY

One search box. Clear answers.

Search, structured medicine cards, scores, charts, comparison and a safety-aware chatbot, all in a single page that runs in any browser.

Thank you

General information only, not medical advice. Always consult a doctor or pharmacist.